Venkata Lakshmi Parimala Pasupuleti

Applied Scientist · Fraud Detection & Prevention · Anomaly Detection · Adversarial ML · Deep Learning · LLMs & GenAI · Python · PyTorch · AWS

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PROFESSIONAL SUMMARY

Applied Scientist with 8+ years building ML models for fraud detection, anomaly detection, and risk modeling. At Entrata, owned the income-limit eligibility pipeline for 7 years: 1.6M+ HUD/Tax Credit records, behavioral and transactional data analysis detecting fraudulent housing applications, zero violations. At George Mason: Isolation Forest anomaly detection across 3 research pipelines; HuggingFace Transformers + PyTorch deep learning training and optimization (full lifecycle). GenAI: adversarial simulation + synthetic data generation (AgentProbe), LLM fraud pattern discovery (LeaseI). M.S. Data Analytics Engineering, GMU (GPA 3.87). — github.com/venkatalakshmiparimala/AgentProbe

WORK EXPERIENCE

Blue Horse Digital
Sep 2026 – Present

AI/ML Engineer (Volunteer) · Remote

- Building ML models and data analysis pipelines for business applications; feature engineering, model training and optimization, and evaluation across behavioral and transactional datasets; applying GenAI and LLM techniques to surface detection strategies from raw business data.

Magizhs Tech Solution
May 2025 – Aug 2025

AI/ML Engineer Intern · Remote

- Built a GenAI adversarial simulation and risk modeling system (LangGraph): parsed inputs into structured risk assessments, generated synthetic adversarial test cases to stress-test detection boundaries against evolving fraud tactics, applied LLM pattern recognition to flag emerging threats; non-compliant scenarios produce grounded rejection reasoning with cited evidence.

George Mason University
Aug 2024 – Aug 2025

Research Associate · Virginia

- Applied Isolation Forest for anomaly detection across 3 research pipelines (surfacing adversarial behaviors manual review missed); built multi-source ML prediction models (Python, R) across 190+ countries ($R^2 = 0.977$); fine-tuned a custom deep learning classification architecture with HuggingFace Transformers + PyTorch (architecture selection, training loop, numerical optimization, evaluation metrics, inference API deployment); presented methodology to faculty.

Entrata (PropTech / Real Estate SaaS)
Sep 2018 – Aug 2024

Senior Software Engineer · Remote

- **ML models for fraud detection and prevention at scale** — owned income-limit eligibility determination models for 7 years: 1.6M+ HUD/Tax Credit records, behavioral and transactional data analysis, historical dataset analysis, anomaly detection, and statistical reconciliation each release cycle. Every release tested against detection thresholds with rollback preparation — a wrong determination is a fraud failure. Zero compliance violations, zero fraud-related security incidents across 7 years. Scalable ML solutions on AWS (S3, EC2, RDS, Lambda, SageMaker); Docker + Kubernetes.
- **Pattern recognition, risk modeling, and data mining** — data mining pipelines to surface risk signals in multi-tenant financial data; 200+ SQL optimizations (30% gain); RabbitMQ async with idempotency; Libsodium PHI-compliant pipelines; 300+ code reviews; 120+ Agile sprints.

PROJECTS

AgentProbe: Synthetic Data Generation, Adversarial Simulation & Model Robustness Testing
LLM-AS-JUDGE · CI REGRESSION

Python · FastAPI · LangGraph · ChromaDB · GitHub Actions · OpenTelemetry | github.com/venkatalakshmiparimala/AgentProbe

- **Synthetic data generation** for adversarial test sets (model robustness against adversarial behaviors and emerging failure modes); LLM-as-judge classifiers; 5-metric eval engine (correctness, groundedness, latency, hallucination rate, token efficiency); **CI regression harness** detecting performance degradation pre-production; multi-provider LLM routing; offline + online evaluation loops — evaluated 1,240+ fraud-sensitive compliance detection test cases.

CrashClarity: Large-Scale Fraud Pattern Detection & Risk Classification (104K Records)
RANDOM FOREST · FEATURE ENGINEERING

Python · R · scikit-learn · Random Forest · Logistic Regression · Feature Engineering | portfolio

- End-to-end ML pipeline on 104K+ records: feature engineering across behavioral variables, Random Forest + Logistic Regression for risk classification (85%+ accuracy), large-scale data mining to uncover hidden risk patterns and detection strategies — full modeling lifecycle: ingestion → feature engineering → training and optimization → evaluation metrics → rapid prototype → dashboard.

AI on FHIR: NLP Pattern Extraction & Behavioral Data Analysis
BERT & EMBEDDINGS · NLP

Python · Flask · LLMs · BERT · spaCy · Embeddings · React · Docker | github.com/venkatalakshmiparimala/AI_on_FHIR

- NLP platform using LLMs, BERT fine-tuning, and embeddings for pattern extraction across large-scale historical records; HIPAA-compliant RBAC; ~60% reduction in manual retrieval time — BERT + semantic search applied to behavioral and historical dataset analysis, analogous to fraud pattern discovery across transaction histories.

EDUCATION

M.S. Data Analytics Engineering George Mason University, Virginia — May 2026

GPA: 3.87

B.Tech Computer Science & Engineering JNTU Anantapur — 2017

SKILLS

ML & Fraud Detection: PyTorch, TensorFlow, Scikit-learn, XGBoost, Isolation Forest, Random Forest, Logistic Regression, Anomaly Detection, Fraud Detection, Fraud Prevention, Risk Modeling, Adversarial Machine Learning, Adversarial Simulation, Adversarial Behaviors, Emerging Threats, Detection Strategies, Model Robustness, Pattern Recognition, Behavioral Data Analysis, Transactional Data Analysis, Data Mining, Large-Scale Data Analysis, Deep Learning Training and Optimization, Transformer Architectures, Feature Engineering, Statistical Modeling, Evaluation Metrics, Scalable ML Solutions, Rapid Prototyping, Model Calibration, Hypothesis Testing, BERT, Hugging Face, spaCy, NumPy, Pandas, SciPy, MLflow, Model Deployment, Model Serving

GenAI & LLMs: LangGraph, LangChain, LLM Evaluation, LLM-as-Judge, Synthetic Data Generation, Adversarial Simulation, Hallucination Detection, RAG Pipelines, Embeddings, Semantic Search, Prompt Engineering, OpenAI API, Anthropic API, Groq, Ollama, LangSmith, ChromaDB, CI Regression Harnesses, Golden Datasets, Red-Teaming, OWASP LLM Top 10, Prompt Injection Defense, Multi-Provider LLM Routing, Agent Observability

Languages & Infrastructure: Python (8+ yrs), Java (Spring Boot), Kotlin, R, TypeScript, JavaScript, SQL, Bash — AWS (S3, EC2, RDS, Lambda, Bedrock, SageMaker), Kubernetes, Docker, PostgreSQL, FastAPI, GitHub Actions CI/CD, OpenTelemetry, Prometheus, Grafana, Distributed Systems, Parallel Computing, PHI Compliance, HIPAA, SOC 2